

Deterministic Finite Automata

FINITE AUTOMATA

A finite automaton is a simple model of a computer

- it is a language recognition device
- theory of finite automaton can be applied in the construction of the lexical analyzer component of a compiler

Deterministic Finite Automaton (DFA)

- Simple language recognition device
- Operation of a DFA is completely determined by the input
- Strings are fed into the device by means of an input tape, which is divided into squares with one symbol inscribed in each square
- Finite control – can sense what symbol is written at any position on the input tape by means of a movable reading head.

Definition:

A deterministic finite automaton (DFA) is a quintuple

$M = (K, \Sigma, \delta, s, F)$ where

K is a finite set of states,

Σ is an alphabet,

$s \in K$ is the initial (start) state,

$F \subseteq K$ is the set of the final states; and

δ , the transition function, is a function from

$K \times \Sigma$ to K

The rules in which the automaton M picks the next state are encoded into the transition function

Example:

DFA $M = (K, \Sigma, \delta, s, F)$

$K = (q_0, q_1)$

$\Sigma = \{a, b\}$

$s = q_0$

$F = \{q_0\}$

δ is defined as

q	σ	$\delta(q, \sigma)$
q_0	a	q_0
q_0	b	q_1
q_1	a	q_1
q_1	b	q_0

A configuration of a DFA is determined by the current state and the unread part of the input string. A configuration is any element of $K \times \Sigma^*$

- $(q_1, w_1) \vdash M (q_2, w_2)$ iff $w_1 = \sigma w_2$ for some symbol $\sigma \in \Sigma$ and $\delta(q_1, \sigma) = q_2$
- $\vdash^* M (q_1, w_1) \vdash^* M (q_2, w_2)$ after 0 or more steps
- A string $w \in \Sigma^*$ is said to be accepted iff there is a state $q \in F$ such that
- $(s, w) \vdash^* M (q, e)$
- Finally, the language accepted by M , denoted by $L(M)$ is the set of all strings accepted by M .

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A DFA can be represented by state transition diagram.

A state diagram is a directed graph. States are represented by nodes, and there is an arrow labeled with σ from node q to node q_1 whenever $\delta(q, \sigma) = q_1$. Final states are indicated by double circles and the initial state is shown by $\triangleright$.

State transition diagram of a DFA

- There is exactly one transition corresponding to each element of Σ , one each state in K .
- Every arc is labeled using one and only one element in Σ .